

Shreya Mahajan

Santa Clara, CA | sm.shreyamahajan@gmail.com | +1 (669)-230-9554 | LinkedIn | GitHub

Full-Stack Software Engineer | Backend, Data & AI Systems

Computer Science graduate and AWS Certified Solutions Architect with hands-on experience building and deploying end-to-end web, backend, data, and AI applications using Python, FastAPI, React, Next.js, TypeScript, SQL, PostgreSQL, and AWS. Built real-time APIs, authentication workflows, analytics dashboards, and cloud data pipelines, including an award-winning platform selected by its sponsor for continued deployment.

Education

Santa Clara University (Santa Clara, CA)

Master of Science in Computer Science and Engineering

Sept 2024 - June 2026

University of Mumbai (Mumbai, India)

Bachelor of Engineering in Information Technology, Honors in Artificial Intelligence and Machine Learning

June 2020 - May 2024

Certifications

AWS Certified Solutions Architect - Associate (SAA-C03), Score: 835/1000

Nov 2025

- Validated AWS knowledge across secure architecture, cost optimization, high availability, IAM, monitoring, compute, storage, and networking.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL

Frontend: React, Next.js, TypeScript

Backend & APIs: FastAPI, REST APIs, WebSockets, Google OAuth, JWT, role-based access

Databases & Data: PostgreSQL, MySQL, DynamoDB, Supabase, vector search

Cloud & Tools: AWS Bedrock, Lambda, S3, EC2, Amplify, Rekognition, IAM, Git, Docker, Vercel

AI/ML: RAG, embeddings, AI agents, NLP, recommendation systems, predictive modeling

Projects

Studentlytics - AWS GenAI Full-Stack Platform (Live Demo)

Oct 2025

- Built an end-to-end AWS GenAI platform for engagement scoring, attendance insights, and chatbot analytics, winning **1st Place HighView Prize among 254 participants** and reducing instructor effort by **80%**.
- Selected by sponsor company HighView for continued development and deployment following the hackathon.
- Built a React and TypeScript analytics dashboard hosted on AWS Amplify and integrated it with FastAPI backend services deployed on EC2.
- Developed AWS Bedrock chatbot workflows using DynamoDB engagement data, S3 storage, Lambda data workflows, and Rekognition-based attendance detection.
- Implemented Google OAuth and JWT-based role access for multiple user types, securing dashboard and chatbot workflows.

Solum - Real-Time AI Agent Integration Platform (Live Demo)

Feb 2026

- Built and deployed a React, Next.js, and TypeScript interface on Vercel that enabled users to select AI personas and initiate real-time voice calls through Twilio Voice and ElevenLabs APIs.
- Implemented a WebSocket-based voice streaming pipeline with measured **sub-200ms response latency** across Twilio Voice, ElevenLabs, and backend agent workflows.
- Built a Supabase/PostgreSQL vector-backed memory layer for retrieval across sessions, enabling personalized and context-aware AI agent responses.

FitGeek - Recommendation and Predictive Analytics System

Jan 2024 - Apr 2024

- Built a Java/MySQL recommendation system using content-based filtering and KNN to generate personalized workout and diet plans, winning **Best Project Award** among **30+ teams**.
- Developed SVM-based diabetes prediction and stress-level detection features, achieving **88% accuracy** on the Kaggle PIMA dataset.

Experience

Graduate Research Assistant, Leavey School of Business (Santa Clara University)

Oct 2025 - June 2026

- Developed Python and Excel data workflows to collect, validate, match, and transform CSV-based regulatory filing and venture capital datasets for regression analysis.
- Documented reproducible methodologies and delivered structured, high-quality analysis with minimal supervision.

Graduate Teaching Assistant, Leavey School of Business (Santa Clara University)

Jan 2026 - June 2026

- Supported approximately **80 students across two academic quarters** through office hours, technical guidance, assignment evaluation, and debugging support.
- Proctored examinations and evaluated coursework, providing constructive feedback on software, data, and machine learning implementations.

Publications

- FitGeek: ML-Based Recommendation System for Fitness and Wellness - ICACEBD-24, 2024**
- Genetic Disorder Prediction using KNN - IJFMR, 2023**